AQ9.3: Activity Questions 3 - Not Graded

This assignment will not be graded and is only for practice.

Directional Derivatives:

Level 1:

1 point

Which of the following statements is(are) true?

- [ ] The directional derivative of $f(x, y)$ at a point (a, b) in the direction of a unit vector u is a non negative real number.
- [ ] The directional derivative of any function is always a scalar.
- [ ] The rate of change of $f(x_1, x_2, ..., x_n)$ in the direction of a unit vector $u = (a_1, a_2, ..., a_n)$ is called the directional derivative of f in the direction of u.
- [ ] The directional derivative of any function is a vector because it is dependent on the direction of a unit vector.

No, the answer is incorrect.

Score: 0

Accepted Answers:

The directional derivative of any function is always a scalar.

The rate of change of $f(x_1, x_2, ..., x_n)$ in the direction of a unit vector $u = (a_1, a_2, ..., a_n)$ is called the directional derivative of f in the direction of u.

If the directional derivative of $f(x, y) = x + xy - 2y^2$ at the point (2,3) in the direction of the vector $(2\sqrt{2}, 2\sqrt{2})$ is $a\sqrt{2}$ where $a \in R$, then find the value of a.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) -3

1 point

If the directional derivative of $f(x, y, z) = xe^y + ye^z + ze^x$ at point (1,0,1) in the direction of a vector (2,-2,1) is $k.e$ where $k \in R$, then find the value of $6k$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 2

1 point

1 point

Suppose f is a multivariable function defined on domain R^2. Which of the following is equal to the rate of change of f at point (x_1, y_1) in the direction of a vector $u = (u_1, u_2)$

- ( )

$\lim_{h \to 0}\frac{f((x_1, y_1) + h(u_1, u_2)) + f(x_1, y_1)}{h}$

○ $\lim_{h \to 0}\frac{f((x_1, y_1) + h(u_1, u_2)) - f(x_1, y_1)}{h}$

○ $\lim_{h \to 0}\frac{f((x_1, y_1) + h(\frac{u_1}{\sqrt{u_1^2 + u_2^2}}, \frac{u_2}{\sqrt{u_1^2 + u_2^2}})) - f(x_1, y_1)}{h}$

○ $\lim_{h \to 0}\frac{f((x_1, y_1) + h(\frac{u_1}{\sqrt{u_1^2 + u_2^2}}, \frac{u_2}{\sqrt{u_1^2 + u_2^2}})) + f(x_1, y_1)}{h}$

No, the answer is incorrect.
Score: 0
Accepted Answers:
$\lim_{h \to 0}\frac{f((x_1, y_1) + h(\frac{u_1}{\sqrt{u_1^2 + u_2^2}}, \frac{u_2}{\sqrt{u_1^2 + u_2^2}})) - f(x_1, y_1)}{h}$

If the directional derivative of $f(x, y) = x\sin(y)$ at the point (1,0) in the direction of a unit vector (u_1, u_2) $(u_1, u_2 \in \mathbb{R})$ is 0.8, then find the absolute value of u_1. (Enter your answer upto one decimal place only)

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Range) 0.5,0.7
1 point

Level 2:

Consider the function $f: \mathbb{R}^2 \longrightarrow \mathbb{R}$ defined as: $f(x, y) = \begin{cases} \frac{x^2 y}{x^4 + y^2} & x, y \neq 0 \\ 0 & x = y = 0 \end{cases}$ Find the directional derivative of f at (0,0) in the direction of the vector $u = (\frac{\sqrt{3}}{2}, \frac{1}{2})$. (Enter your answer upto one decimal place only)

No, the answer is incorrect.
Score: 0

Accepted Answers:
(Type: Range) 1.4,1.6
1 point
Suppose dd is the directional derivative of a scalar valued multivariable function $f(x, y) = x^2 + y^2$ f(x, y) = x2 + y2 at a point $P(1, 2)$P(1, 2) in the direction of the line PQPQ, where point QQ is at (4,6). The value of dd is (Enter your answer upto one decimal place only)

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Range) 4.3,4.5
1 point
Suppose $f(x, y, z) = ax^2 y + yz^2 - z^2 x$f(x, y, z) = ax2y + yz2 − z2x, where $a \in R$a ∈ R, is a scalar valued multivariable function defined on domain $D \subset R^3$D ⊂ R3. For what value of aa, does the directional derivative of the given function at the point (1,1,1) in the direction of the vector $u = (1, 2, -2)$u = (1, 2, −2) equal 33?

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 2
1 point
The temperature at a point (x, y, z) is given by

$$T(x, y, z) = 10e^{-x^2+y^2-4z^2}$$

where T is measured in $^\circ C$ and x, y, z are measured in meters. If the rate of change of temperature at (2,-1,0) in the direction of a vector $u = (1, -1, 1)$ is $k.\frac{e^{-3}}{\sqrt{3}}$ where $k \in \mathbb{R}$, then the value of k is (approximate to two decimal places)

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) -20
1 point
Let $f : R^2 \to R$f : R2 → R be given by $f(x, y) = x^2 + y^2 - 2xy$f(x, y) = x2 + y2 − 2xy. Suppose $D_u f(x_0, y_0)$Duf(x0, y0) is the directional derivative at (x_0, y_0)(x0, y0) in the direction of the unit vector uu = $(\frac{1}{\sqrt{2}}, \frac{-1}{\sqrt{2}}$ 2

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−1) and $D_v f(x_0, y_0)$Dvf(x0, y0) is the directional derivative at (x_0, y_0)(x0, y0) in the direction of the unit vector vv = $(\frac{1}{2}, \frac{\sqrt{3}}{2}$21, 2 3

√

). Consider a matrix $A = \begin{bmatrix} D_u f(1,2) & D_u f(2,1) \\ D_v f(1,2) & D_v f(2,1) \end{bmatrix}$A = [Duf(1, 2)Dvf(1, 2)Duf(2, 1)Dvf(2, 1)]. What is the value of $det(A)$det(A)?

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 0
1 point

Check Answers

Your score is: 0/10